

Progress Report: Five Grokking Predictors on the 5-Seed Nanda-Unified Run

Thesis: *A Unified Benchmark of Grokking Predictors in Neural Networks*

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Overview

Today the 5-seed baseline run of the Nanda-Unified protocol was completed and five grokking predictors (L2 Norm, Dropout, Spectral, AGE and HTSR Alpha) were computed and scored against the grok epoch. **Main result: none of the five works as implemented.** All code, results, checkpoints and this report are committed in the repository (08_Experiments/results/nanda_unified/, 06_Code/, context.md).

1. Objectives completed

- Finish the 5-seed training run of the Nanda-Unified protocol (modular addition, $p = 113$, 40000 epochs). All 5 seeds grokked.
- Compute Predictors 1 to 4 (L2 Norm, Dropout, Spectral, AGE) on all 5 seeds and score them with one common test.
- Implement Predictor 5 (HTSR Alpha) following Martin and Mahoney, compute it on all 5 seeds and score it with the same test.
- Record the scope decision: predictors are evaluated with their published methods; no new detection rule is introduced.

2. Work accomplished

Protocol. 1-layer transformer, 4 heads, $d_{\text{model}} = 128$, no LayerNorm, small initialisation $N(0, 0.8/\sqrt{d_{\text{model}}})$, AdamW (learning rate 0.001, betas 0.9 and 0.98, weight decay 1.0), full batch, 30/70 train/test split, Apple Silicon (MPS). Grok epoch = first epoch with test accuracy above 0.9. About 2.1 hours per seed.

Seed	0	1	2	3	4	Mean (std)
Grok epoch (test acc > 0.9)	12987	6921	9510	11019	23747	12837 (5804)

- **Checkpoints:** weights saved every 50 epochs (801 per seed, about 3.2 GB in total), so Predictors 3 to 5 are computed from frozen checkpoints without retraining.
- **Pipeline clean-up:** runner path bug fixed (broken since the 17 September reorganisation); arbitrary constants moved from the runner into the configuration file with their sources.
- **Spectral and AGE** computed from the checkpoints (about 1 hour per seed for Spectral, seconds for AGE).
- **HTSR Alpha** implemented in a new predictor file: for each of the 6 linear weight matrices, the eigenvalues of $W^T W / N$ are computed and the power-law exponent α of the tail is fitted by maximum likelihood, with the tail start chosen automatically by the smallest Kolmogorov-Smirnov distance (Clauset, Shalizi and Newman 2009; *powerlaw* package). The summary signal is the mean α over the 6 layers. No hand-picked threshold is used.
- **Analysis:** one analysis script scores all five predictors against the grok epoch and produces the figures below.
- **Shared model file and earlier predictor code were not changed.**

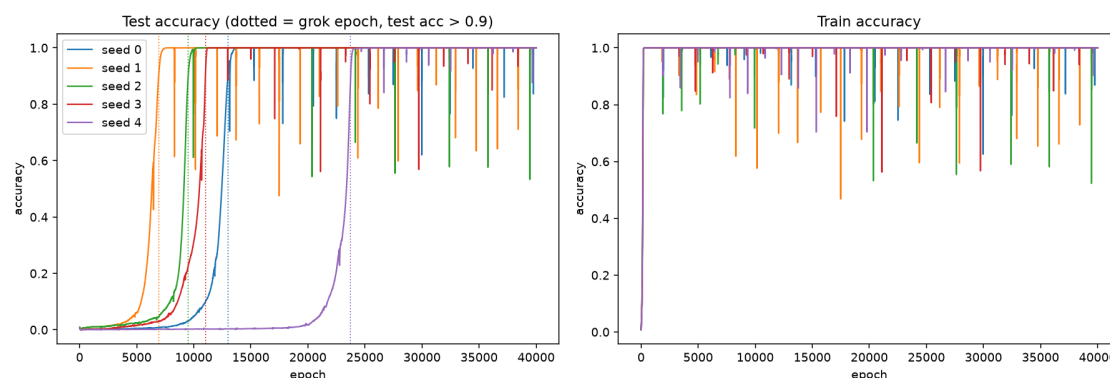


Figure 1. Test and train accuracy for the 5 seeds (dotted lines mark the grok epochs). The brief drops after grok are discussed in Section 3.

3. Challenges encountered

- **Spectral and AGE were empty after training.** Cause: the runner's default predictor list excluded them (not a code bug). Fixed by re-running on the saved checkpoints, with no retraining.
- **Disk space:** the disk was 99 percent full before the 3.2 GB of checkpoints could be committed; space was freed first. Pushing to the remote will be slow because of the checkpoint size.
- **HTSR alpha below 2.** The fitted alpha is below 2 in 85 to 94 percent of layer-checkpoint fits, whereas the well-trained range in Martin and Mahoney's interpretation is roughly 2 to 6. The layers are small (at most 128 eigenvalues) and the automatic tail start sometimes keeps as few as 4 eigenvalues, so the method is operating partly outside its designed range. Values are reported as computed; nothing was tuned.
- **Brief collapses of accuracy after grok (Figure 1).** After grokking, test accuracy drops for about 2 to 10 epochs at irregular intervals and then recovers to 1.0. Train accuracy drops at the same time (in seed 1 it falls to 0.47), so the model briefly breaks on its own training data; this is a training instability, not overfitting. The number of post-grok epochs below 0.9 test accuracy is 28, 75, 31, 18 and 4 for seeds 0 to 4, and no seed shows a sustained cycle. The likely cause is the interaction of full-batch AdamW with weight decay 1.0: once the loss is almost zero, the gradients are almost zero, weight decay keeps shrinking the weights, and Adam's rescaling of small gradients then produces a sudden large update (the loss jumps from about $1e-5$ to between 0.4 and 3.5, then recovers). This is consistent with the known slingshot effect of Adam-type optimisers (Thilak et al. 2022). It is **not proven** here: no run with a different learning rate or weight decay was done, and the weight norm was lower at the dips in seed 1 but not in seed 4. The grok epoch, the final accuracies and the predictor verdicts are not affected, because all collapses come after the grok epoch.
- **Statistical power:** with 5 seeds only a rank correlation of 0.9 or more is significant, and seed 4 (latest grok) dominates several correlations.
- **Timing resolution:** the evaluation grid is 100 epochs, so timing finer than 100 epochs cannot be resolved.

4. Results obtained

Scoring. Test 1: the event must occur *before* the grok epoch in all 5 seeds (lead = grok epoch minus event epoch). Test 2: the event epoch must follow the grok epoch across seeds (Spearman rho; with $n = 5$, $\rho = 0.9$ gives p of about 0.037 and $\rho = 1.0$ gives p of about 0.017).

Predictor and rule	Event epoch, seeds 0 to 4	Before grok	rho
L2: moving-average crossover	113 / 113 / 120 / 130 / 136	5/5	0.70
L2: MA-of-MA zero crossing	1802 / 1767 / 1814 / 1810 / 1801	5/5	0.00
Dropout: variance peak	24200 / 25400 / 16700 / 18800 / 39000	0/5	0.40
Spectral: k90 minimum	29700 / 30700 / 14100 / 38000 / 33900	0/5	0.30
Spectral: alignment maximum	31600 / 37800 / 29300 / 37400 / 32000	0/5	-0.30
AGE: NC1 minimum	31500 / 37800 / 29300 / 37400 / 30100	0/5	-0.40
HTSR: mean alpha minimum	34300 / 16000 / 24100 / 19500 / 39600	0/5	0.90

Table 1. Predictors as implemented. Grok epochs are 12987 / 6921 / 9510 / 11019 / 23747.

- **L2 Norm** fires before grok in all seeds but at almost the same epoch every time (113 to 136, or about 1800), while grok moves from 6921 to 23747. It detects an early training transient, not grokking.
- **Dropout, Spectral, AGE and HTSR** reach their extreme only after grok, in every seed. The extreme lies in the noisy post-grok phase or on a flat plateau, so the reported epoch is late and partly arbitrary.
- **HTSR Alpha**: mean alpha falls from about 2.7 to about 1.25 in every seed and is already about 1.5 at the grok epoch. Its rho of 0.90 is borderline: seed 4 drives it, and Test 1 already fails.
- **Effect of the collapses on the predictors**: the extreme values of Dropout variance, Spectral alignment and AGE NC1 fall inside this noisy post-grok phase, which is a likely reason why the extreme-value reading arrives late.
- **The signals are informative; the published extreme-value reading is what fails.** In an illustrative retrospective check (arbitrary 10 and 50 percent levels, not a benchmark result and not a proposed rule), the 50 percent points of Dropout variance, Spectral alignment and AGE NC1 lie within about 800 epochs of grok and rank the seeds correctly ($\rho = 1.00$). This check was not repeated for HTSR.

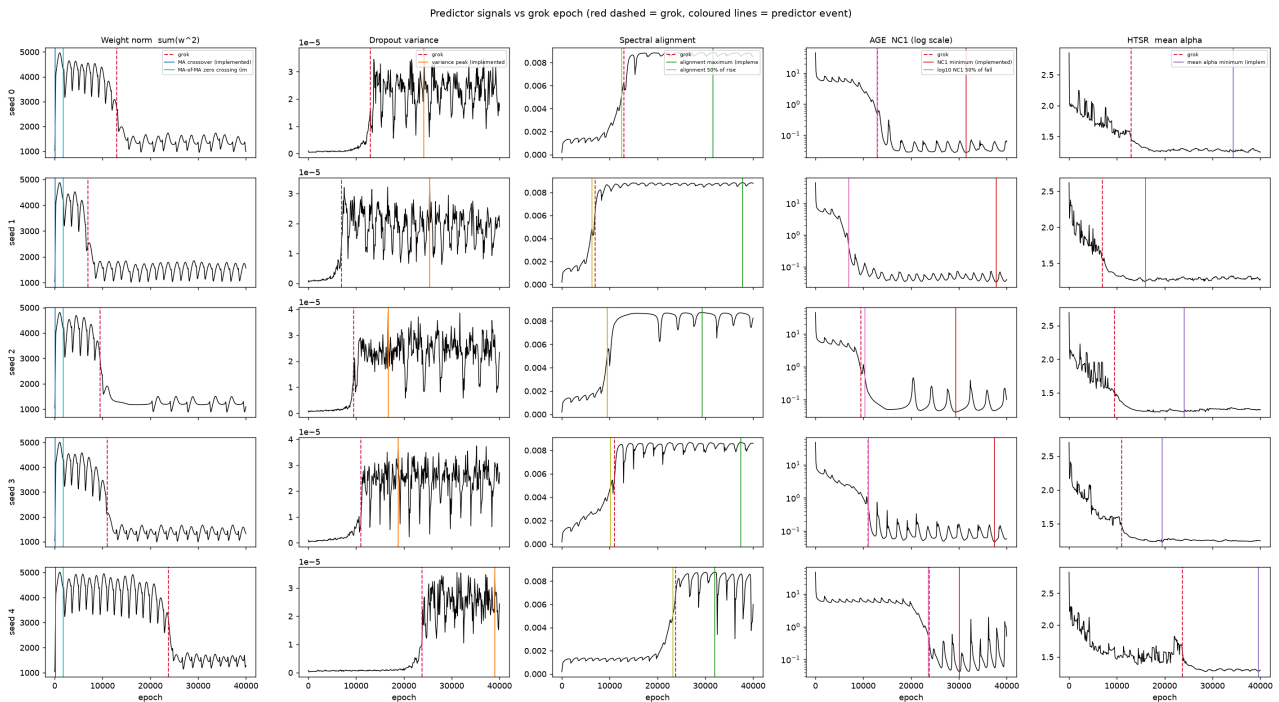


Figure 2. Raw predictor signals per seed (rows). Red dashed line: grok epoch. Coloured lines: each predictor's event epoch. The signals change around grok; the events fall later.

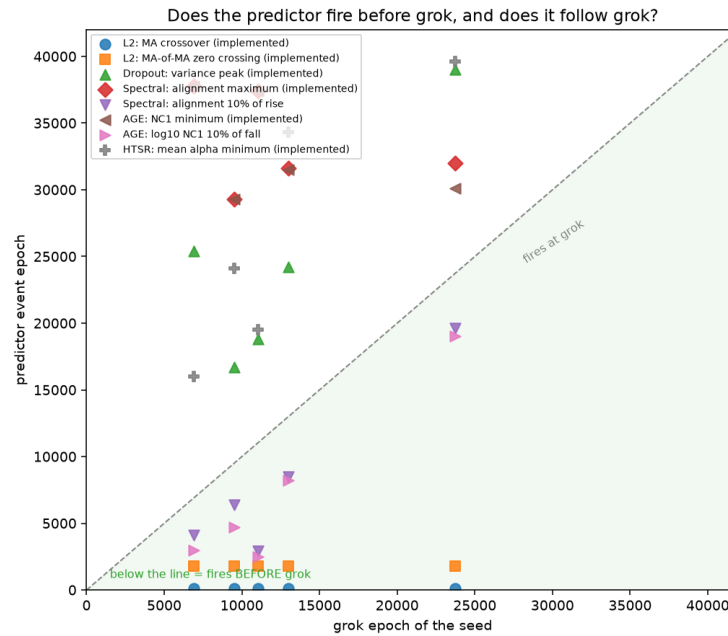


Figure 3. Event epoch against grok epoch. Points below the diagonal would mean the predictor fires before grok.

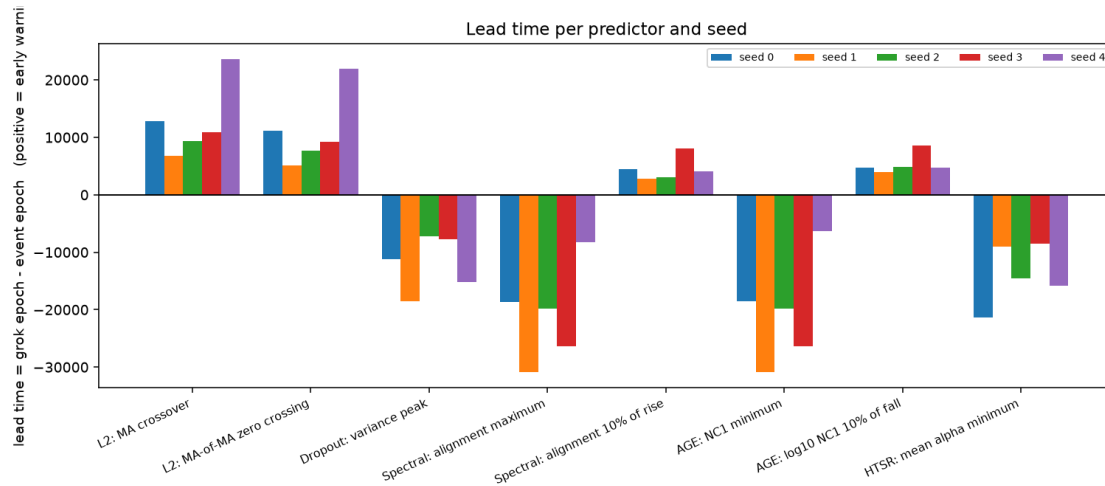


Figure 4. Lead time (grok epoch minus event epoch). Negative bars mean the predictor fired after grok.

5. Plans for the next reporting period

- Continue the evaluation order: **Correlation Traps** (Predictor 6), then Weight-PCA, Higher-MI and Commutator Defect, each with its original paper's method and the same scoring test.
- Extend the analysis script to every new predictor, so all nine are compared on the same 5 seeds.
- Start writing the results chapter for Predictors 1 to 5 in 05_Thesis_Document.
- **For discussion in the supervision meeting:** (1) how to present the HTSR alpha below 2 (report as computed with the caveat, or add per-layer alpha, fit-quality and tail-size plots); (2) whether the retrospective sensitivity check should appear in the thesis at all, given the scope decision not to develop a new detection rule; (3) whether 5 seeds are enough or more seeds should be added for the final benchmark; (4) whether to run one extra seed with a lower learning rate or weight decay, to test whether the post-grok collapses disappear.

Limitations recorded: single task and architecture (modular addition, $p = 113$, 1-layer transformer); no online detection rule is developed, so "no predictor is proven usable" is a documented limitation and future work.